AI for Empowering Women in AI

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Abstract

Women remain significantly underrepresented in computing and artificial intelligence (AI), facing barriers such as limited access to tools, training, and opportunities. As AI becomes increasingly integral to daily life, it has the potential to address these disparities and foster greater inclusion. Realizing this potential requires fair access, inclusive design, and strategies that actively promote confidence, participation, and representation. This study explores these dimensions, identifying pathways for AI to serve as a catalyst for change.

A key finding is the lack of tools explicitly designed to support women's participation in AI. Beyond reducing bias, fostering diversity in AI development requires designing artifacts that actively promote knowledge, skills, and confidence. Empowerment in this context refers to equipping individuals to engage with and influence AI, whether as users or developers. Women's participation in AI can thus be understood as a cyclical process, where increased engagement leads to further inclusion and representation in the field.

The research objective of this study is to provide information on the relation between empowerment and women in AI, using the Systematic Literature Review (SLR) methodology. The SLR proposes and analyzes 14 studies that serve as a basis to understand the relationship between women, empowerment, and AI.

The results show that there is still scarce research explicitly connected to AI artifacts designed for empowerment, but there is a growing recognition of its importance. The SLR also reveals various challenges and success factors related to the role of AI in fostering gender inclusivity and empowerment, which calls for further attention.

As further work, the authors will conduct empirical research to validate the findings and gather new insights, while co-designing an artifact to empower women in AI.

CCS Concepts

• **Social and professional topics** → **Computing / technology policy**; **Computing industry**; **Computing and business**; • **Human-centered computing**; • **Computing methodologies** → **Artificial intelligence**;

*Vestvik: query, paper analysis, writing. Jaccheri: project idea, supervision.

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Keywords

AI, Empowerment, Women, Diversity, Inclusion, Computer Science, Systematic Literature Review

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1 Introduction

Historically, women were pioneers in computing, and in the 1940s, they made up the majority of the workforce [33]. However, as the field gained prestige, they were systematically pushed out, leading to a significant gender gap that persists today.

Limited access to AI training and tools, along with lower confidence levels, further reinforces this underrepresentation [26]. Women are less likely to be offered opportunities to upskill in AI and report lower usage of AI for problem-solving at work [26]. These disparities highlight the urgent need for targeted initiatives that expand training access, promote confidence, and ensure equitable opportunities [26].

As AI continues to become a bigger part of everyday life, it is increasingly important to ensure that women are both contributors to the field of AI and equipped to understand and utilize its applications. AI systems are trained on datasets selected by humans, and if these datasets contain biases, the AI will inevitably reflect and perpetuate these biases [2]. This shows the critical need for diversity in AI development. A diverse and inclusive approach to AI development is crucial, as a representative group of developers and decision-makers can better identify and mitigate biases that might otherwise go unnoticed.

The significant underrepresentation of women in AI serves as a key motivation to explore how they can be empowered in the field. The emergence of generative AI, exemplified by ChatGPT in late 2022, marks a significant shift in how AI is perceived and utilized [1, 29]. Despite rapid adoption of AI [1], research on how different groups interact with AI and feel empowered by it remains limited. This gap underscores the need for tools and strategies that not only enhance AI literacy but also empower individuals to use AI effectively. To bridge this gap, it is essential to develop tools and strategies that enhance women's technical skills and confidence in AI, ensuring their active participation in the field.

The objective of this research is to explore existing knowledge and studies on women's empowerment in AI, with a particular focus on their participation, challenges, and contributions in the field. The project seeks to understand how AI can foster empowerment

for underrepresented groups, identify key characteristics of an artifact that can achieve this goal, and uncover gaps in the existing literature. By addressing these challenges, this research aims to identify key strategies for leveraging AI as a tool for empowerment. Accordingly, this study explores the following research questions:

- **RQ1**: How do recent studies address the empowerment of women in AI?
- **RQ2**: What are the characteristics of an AI artifact that facilitates the empowerment of women?

To address the research questions, a Systematic Literature Review (SLR) was performed to identify gaps and gain insights into relevant areas of the primary studies. The methodology will follow the principles outlined by Kitchenham [10].

The structure of this paper is organized as follows: Section 2 provides the background, offering insights into the field of AI, gender equality, bias, and diversity within computer science. Section 3 describes the research methodology, focusing on the SLR. The SLR results are presented in Section 4, followed by a discussion of the findings and their implications in Section 5. Section 6 concludes the paper.

2 Background

2.1 Gender Equality & Artificial Intelligence

Achieving gender equality, as defined by the United Nations Sustainable Development Goal (SDG) 5, requires targeted actions across all areas of knowledge [22]. AI has become an important part of modern life and economic activities, playing a crucial role in addressing societal challenges and driving innovation. Its potential to influence both developed and developing nations, makes AI a key enabler of progress across various fields [15].

Significant gender disparities exist in AI engagement in the workplace [19]. Women are less likely to be offered skill development opportunities, feel less confident in their AI skills, and have lower access to AI tools compared to men [19]. Fewer women use AI for problem-solving, and they are also less likely to be provided AI access by their employers [19].

Although AI holds promise for advancing gender equality, its impact is not inherently positive. If implemented irresponsibly, AI systems risk amplifying existing biases [25]. Nevertheless, AI can support SDG 5 by identifying systemic inequalities, reducing biases, and creating new opportunities for women in both industrialized and developing nations [18]. This dual role of AI, serving as both a driving force of empowerment and a potential sustainer of bias, underscores the need for careful design, implementation, and governance to ensure equitable outcomes.

2.2 Diversity in Computer Science

Computer science has one of the largest gender gaps within the fields of science, technology, engineering, and mathematics (STEM) [5].

Stereotypes portraying computer science as male-dominated and requiring exceptional intelligence significantly diminish girls' interest and sense of belonging to the discipline [5]. Women in male-dominated fields, such as computer science, report higher levels of gender discrimination and are more likely to feel threatened by negative stereotypes regarding their abilities. Moreover, women in these fields are more likely to consider changing their major compared to both men in the same fields and women in female-dominated fields [30].

The lack of a sense of belonging is a significant factor contributing to gender imbalance in computing [37]. Fostering a sense of belonging is crucial, as the underrepresentation of women in the field creates a reinforcing pattern where fewer women in computing leads to an even smaller presence over time. These findings underscore the persistent challenges women face in male-dominated academic disciplines and emphasize the need for interventions that address discrimination and stereotype-driven barriers in education and career progression [30].

Intersectionality is a framework that recognizes the interconnected nature of various social identities and systems of oppression, highlighting how individuals may experience unique forms of discrimination and marginalization at the intersection of multiple identities [7]. The computer science field has traditionally been dominated by a homogeneous group, primarily consisting of "white" men, which implies a lack of representation from various dimensions of diversity such as ethnicity, gender, disability, and other factors related to intersectionality [13, 16].

2.3 Empowerment in AI

Empowerment is the process of gaining the autonomy, authority, and confidence to make decisions, take initiative, and exert control over one's own life, work, and circumstances. Empowerment goes beyond the delegation of responsibility; it involves fostering autonomy, self-efficacy, and active participation in decision-making processes [11].

In the context of AI, empowerment involves equipping individuals with the tools and knowledge to understand AI's potential, engage with its applications, and effectively leverage its opportunities. This enables individuals to take control of their lives and achieve personal and professional goals while adapting to an AI-driven world. Empowerment through AI emphasizes providing people with the necessary skills to interact with AI systems, fostering confidence, competence, and greater autonomy [12, 17].

3 Research Method

This study followed the guidelines for a SLR as established by Kitchenham [10], ensuring the identification of all relevant studies related to the research question. SLR was selected to establish a foundational understanding of how the concept of empowerment is currently addressed in AI research. This method enables a structured overview of patterns, gaps, and trends across multiple studies, which is essential for identifying areas where further empirical work is needed.

To complement the systematic approach, additional relevant articles were identified by reviewing studies cited in Patón-Romero's paper State of Gender Equality in and by Artificial Intelligence [23]. Since this study closely aligns with the research topic of [23] , the authors build on the work of [23] and use the same search query in the SLR. This further strengthens the foundation and enhances the depth and relevance of the current analysis.

Table 1: Research Question and Motivation.

Research Question	Motivation
RQ1: How do recent studies address the empowerment of women in AI?	To gain an overview of recent studies addressing empowerment within AI.
RQ2: What are the characteristics of an AI artifact that facilitates the empowerment of women?	To identify the specific design elements and functionalities that make an AI tool accessible, engaging, and empowering for women.

3.1 Identification of Research

The objective of this research is to explore existing knowledge and studies on women's empowerment in AI, with a particular focus on their participation, challenges, and contributions in the field.

"Women in AI" is defined, in this paper, as women who either use or develop AI. This definition highlights the dual relationship between women and AI: women engage with AI by using or developing it, while AI, in turn, has the potential to empower women by providing opportunities, skills, and resources. Figure 1 visualizes this interaction, showing how women contribute to AI and how AI, in return, can support and enhance women's empowerment.

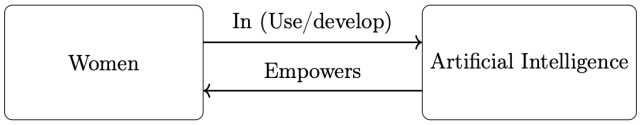

Figure 1: On the Relation between Women and AI

In line with the research objective, the following research questions have been established, as presented in Table 1.

3.2 Data Collection

Scopus was selected as the primary database for this SLR due to its extensive coverage and advanced search functionalities. To build upon existing research, this study adopted the search query from Patón-Romero [23]. This query incorporated various spellings and phrasings of terms such as "AI," "Women", and "Empower" to ensure a comprehensive search. Since the studies from [23] cover research from 2011 up to 2021, the search was restricted to the period between 2022 and 2024, focusing exclusively on topics within Computer Science. Earlier studies were excluded, as [23] had already addressed similar research questions and extensively explored the themes of "Gender in AI" and "Gender by AI", providing comprehensive coverage of prior work. The search query used in this study is presented in Table 2, and was run in late September 2024.

3.3 Selection of Primary Studies

The search query in Scopus returned 242 initial results. Then, the inclusion and exclusion criteria, detailed below, were applied to the titles and abstracts, resulting in the selection of 12 studies. A quality assessment was later implemented, further reducing the final number of studies to 6. The dataset containing the initial results from the Scopus search is available on Mendeley Data [35].

Table 2: Search Query.

Database	Search Query	Hits
Scopus	("Artificial Intelligence" OR AI) AND ((Women OR Girls OR Gender) AND (Equal* OR Empower*))	242

Additionally, the 27 studies identified in [23] were reviewed, and the same inclusion and exclusion criteria were applied. As a result, the selection was narrowed to 16 studies. A quality assessment was later implemented, resulting in a final total of 8 studies, all from 2016–2021, as earlier publications did not meet the criteria. The selection process is illustrated in Figure 2. After completing the selection process, the SLR resulted in 14 primary studies. The dataset containing these primary studies is available on Mendeley Data [35].

The first author was responsible for the process under the supervision of the second author.

Inclusion criteria

(1) The paper must address at least one of the RQs.
(2) The paper should focus on empowerment or inclusion in AI.
(3) The paper should be within the fields of CS or STEM.
(4) The paper should be open access.

Exclusion criteria

(1) The paper is not in English.
(2) The paper is published before 2016.
(3) The paper is duplicate work.
(4) The paper is less than 6 pages in length.
(5) Paper about women and AI outside of CS or STEM.

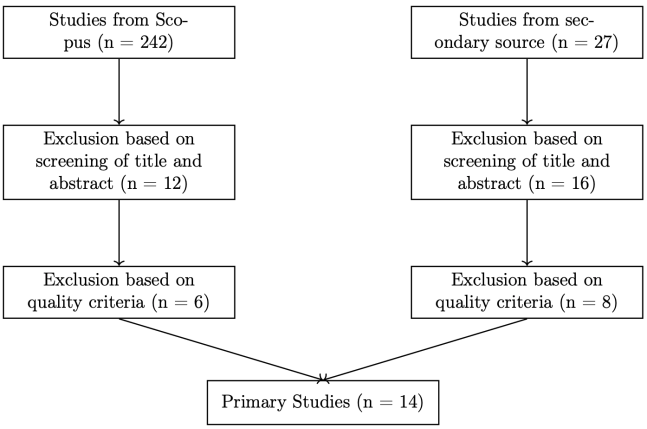

Figure 2: Selection Process used in the SLR.

3.4 Data Extraction and Data Synthesis

The goal of the data extraction process was to summarize the information obtained from the included studies in a systematic and structured way. To achieve this, all relevant data were compiled into a predefined Excel sheet serving as a data extraction form. This form included general study information, as well as key findings related to the research questions. Organizing the data in this way provided a comprehensive overview of the studies and facilitated a consistent assessment of their relevance and alignment with the review questions.

To maintain quality, a consistency check was performed in which the author extracted data from a random sample of studies twice and compared the results to ensure accuracy. The synthesis was conducted descriptively, and the information extracted was organized to align with the review question, ensuring that the data effectively addressed the main focus of the review.

4 Results

An overview of the 14 primary studies is available on Mendeley Data [35], and their titles and aims are summarized in Table 3.

4.1 *RQ1: How do recent studies address the empowerment of women in AI?*

Research about AI and gender has gained increasing attention in recent years, reflecting a broader awareness of gender disparities in the field. The relatively small sample of 14 primary studies illustrated in Figure 3 makes it difficult to identify clear publication trends. A broader analysis of all articles retrieved from Scopus between 2022 and 2024, based on the search query presented in Table 2, provides a clearer picture of the overall research landscape, as shown in Figure 4. The dataset comprised a total of 242 articles, 57 from 2022, 83 from 2023, and 103 from 2024, indicating a noticeable increase in research interest over time. The full dataset of retrieved articles is available on Mendeley Data [35].

The geographical distribution of the studies highlights clear disparities in research contributions. The United States had the highest number of publications with three studies, followed by several European countries contributing a total of eleven. Although Europe dominates, countries in Eastern Europe are notably underrepresented.

Entire continents such as South America, Africa, and Oceania are completely absent from the primary studies. This absence points to a significant geographical gap in the research coverage and suggests that these regions remain underrepresented in the field. Most of the studies originate from countries in the Global North, with China as the only exception. The term "Global North" generally refers to economically developed countries, while the "Global South" encompasses countries with emerging or developing economies [20]. This distribution underscores the lack of research contributions from economically developing region.

The distribution of research strategies reveals that case studies make up the majority, focusing on the impact of specific interventions. Empirical studies, which involve data collection and analysis, represent a smaller portion, followed by theoretical studies offering conceptual insights. Literature reviews and systematic mapping studies each make up an equal share.

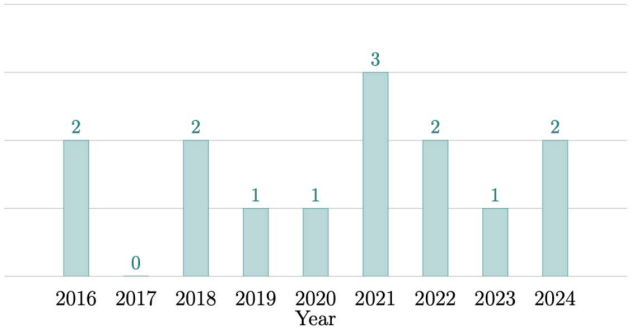

Figure 3: Publication Year distribution for the primary studies.

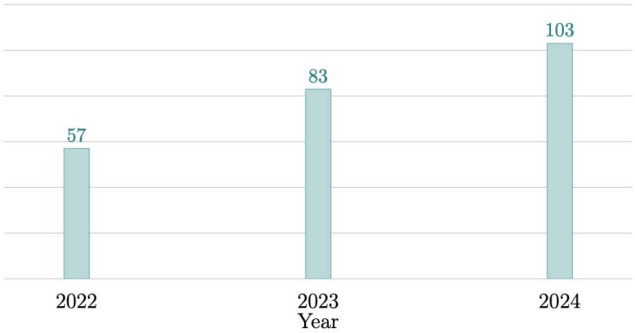

Figure 4: Publication Year Distribution for the full dataset of retrieved articles.

Although all selected studies address gender diversity in AI, relatively few explicitly focus on empowerment in AI. Only one study, S2 [27], includes both "empowerment" and "AI" in its title, and explicitly highlights both as focal points in its objectives. Instead, many studies use alternative terminology, with recurring keywords such as "gender", "women", "diversity", "bias", and "embedding". Some studies, such as S8 [38], explore empowerment indirectly using AI to challenge gender norms, while others, such as S13 [3], examine how word embeddings in AI reflect and potentially amplify societal biases.

The studies can be grouped into three main themes: research on AI and gender (S4 [22], S10 [36], S11 [21]), methods of engagement in AI (S2 [27], S3 [8], S5 [14], S6 [12], S8 [38], S14 [34]), and AI artifacts designed for empowerment (S1 [32], S7 [9], S9 [24], S12 [31], S13 [3]). Studies in the first category primarily examine the relationship between AI and gender, often analyzing biases embedded in AI systems and how they impact societal norms. The second category explores methods to actively engage people in AI-related activities, including interventions designed to increase participation and awareness. Finally, the third category focuses on the development of AI artifacts aimed at empowerment, such as tools designed to mitigate bias or promote inclusivity. Together, these studies illustrate various approaches to addressing gender diversity and empowerment in AI, ranging from theoretical analysis to practical implementations.

Table 3: Titles and Aims of Primary Studies

Study ID	Title	Study Aim
S1 [32]	How Low-Code Tools Contribute to Diversity, Equity, and Inclusion (DEI) in the Workplace: A Case Study of a Large Japanese Corporation	To examine the impact of low-code tools on promoting diversity, equity, and inclusion within corporate environments.
S2 [27]	Empowering Female Founders with AI and Play: Integration of a Large Language Model into a Serious Game with Player-Generated Content	To develop a serious game prototype aimed at empowering female entrepreneurs by raising awareness of gender-specific biases.
S3 [8]	A Multimodal Installation Exploring Gender Bias in Artificial Intelligence	To create an art installation that highlights gender bias in AI, revealing how lack of transparency can exclude women.
S4 [22]	Artificial Intelligence and Gender Equality: A Systematic Mapping Study	To explore the relationship between AI and social sustainability, with a focus on gender equality as a key SDG.
S5 [14]	Robots as My Future Colleagues: Changing Attitudes Toward Collaborative Robots by Means of Experience-Based Workshops	To investigate changes in participants' willingness to collaborate with robots after a workshop.
S6 [12]	Evaluation of an Artificial Intelligence Literacy Course for University Students with Diverse Study Backgrounds	To design and implement a flipped-classroom initiative to teach AI to university students.
S7 [9]	Debiasing Career Recommendations with Neural Fair Collaborative Filtering	To reduce gender bias in career recommendations using fair collaborative filtering techniques.
S8 [38]	Boosting Robot Credibility and Challenging Gender Norms in Responding to Abusive Behaviour: A Case for Feminist Robots	To explore feminist social robots to challenge gender norms and reduce bias.
S9 [24]	Bias in Multimodal AI: Testbed for Fair Automatic Recruitment	To develop a simulated recruitment framework to examine algorithmic gender and racial biases.
S10 [36]	Feminist AI: Can We Expect Our AI Systems to Become Feminist?	To explore the origins of gender and racial bias in AI and evaluate proposed solutions.
S11 [21]	A Gendered Perspective on Artificial Intelligence	To reduce gender imbalance in AI by applying fairness standards and ethical principles.
S12 [31]	Biased Embeddings from Wild Data: Measuring, Understanding and Removing	To examine and address bias in word embeddings related to gender and race.
S13 [3]	Man is to Computer Programmer as Woman is to Homemaker? Debiasing Word Embeddings	To demonstrate how word embeddings reflect and amplify gender stereotypes.
S14 [34]	Toward More Gender Diversity in CS through an Artificial Intelligence Summer Program for High School Girls	To evaluate a program designed to increase girls' interest in AI and reduce barriers to CS education.

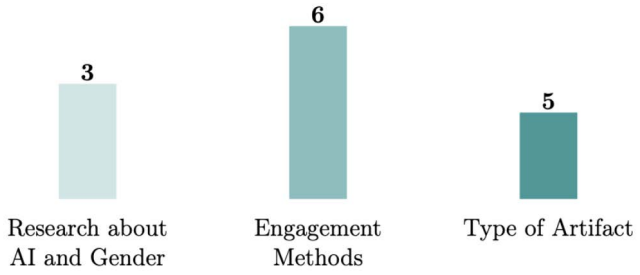

Figure 5: Primary Studies: Number of Articles by Category.

4.2 *RQ2: What are the characteristics of an AI artifact that facilitates the empowerment of women?*

Building on the three categories identified in RQ1, this section examines the characteristics of AI artifacts that contribute to women's empowerment by analyzing how different studies define and implement empowerment through AI.

4.2.1 Research about AI and Gender. The primary studies presented in Table 4, focus on the intersection of AI and gender issues.

S4 identifies significant gender gaps in AI, emphasizing AI's dual role in supporting or hindering gender equality. AI can help address biases and eliminate discrimination if designed with transparency

and inclusivity [22]. However, the study warns that without diversity in development teams, AI risks reinforcing biases, leading to algorithms that fail to address women's needs [22].

S10 and S11 further highlight how AI biases reinforce gender inequalities. S10 stresses that AI inherits biases from training data, requiring active mitigation [36]. S11 illustrates how AI reflects traditional gender roles and underscores the need for diverse teams to create more equitable systems [21].

These findings highlight the role of AI, emphasizing the importance of deliberate development practices to prevent unintentional bias and maximize AI's potential as a tool for gender equality.

Table 4: Articles Focusing on Research About AI and Gender.

Study ID	Analytical Focus	Findings
S4	Intersection of AI and gender equality	Identifies gaps in gender equality within AI, emphasizes the need for research on SDG 5 targets, and highlights how AI can both support and hinder gender equality initiatives.
S10	Potential for AI to embody feminist principles	Explores sources of gender bias in AI systems and suggests four strategies to address bias: omitting gender data, improving transparency, designing non-biased algorithms, and involving humans in decisions.
S11	Gender perspectives in AI development	Analyzes gender norms in AI, calling for more inclusive design processes and representation to mitigate biases in AI technology and enhance gender diversity.

4.2.2 Methods of Engagement. Table 5 presents the primary studies that focus on methods for using AI to empower women, highlighting both the engagement strategies employed and the key findings.

Focusing on methods for using AI to empower women, engagement approaches vary widely, including games, workshops, literacy courses, robotic interactions, and summer programs. S2 and S3 highlight efforts to raise awareness of gender discrimination and equip individuals with response strategies [8, 27]. The workshop in S5 and the robot interactions in S8 demonstrate how AI can change perceptions, fostering a more open-minded attitude [14, 38]. S8 further illustrates how assertive robot behavior challenges gender roles and promotes feminist principles in AI [38].

AI literacy courses and summer programs, as examined in S6 and S14, build foundational AI knowledge and confidence, particularly among underrepresented groups [12, 34]. The summer program, specifically targeted at young women, emphasizes the importance of role models and a supportive network to foster interest in careers in AI and computer science. [34] This approach contributes to breaking down stereotypes and encouraging broader participation in the field of AI.

These findings showcase diverse strategies for empowering women and underrepresented groups by reshaping perceptions, challenging AI bias, and fostering inclusivity.

Table 5: Articles Focusing on AI Engagement.

Study ID	Engagement Method	Findings
S2	Game	Increases resilience, awareness of gender-based discrimination, and provides skills to respond effectively in discriminatory scenarios within the startup environment.
S3	Interactive Installation	Enhances awareness of gender bias in AI, encouraging participants to critically reflect on their role in shaping digital futures and recognize bias in AI technologies.
S5	Workshop	Shifts perceptions of robots from potentially threatening to empowering, fostering a positive attitude toward collaborative robots as future colleagues.
S6	AI Literacy Course	Increases conceptual understanding of AI, particularly in machine learning, empowering students from diverse backgrounds to work with and understand AI.
S8	Robot Interaction	Improves perceptions of robot credibility, challenges gender norms, and empowers users by modeling non-subservient responses to abusive behavior, promoting feminist principles.
S14	Summer Program	Boosts interest and confidence in AI, provides role models, and fosters a supportive community, encouraging high school girls to pursue careers in AI and computer science.

4.2.3 Type of Artifact. Table 6 presents studies that focus on AI artifacts, outlining the features designed to reduce gender bias and support empowerment.

Studies examining AI artifacts focus on their ability to reduce gender bias and contribute to empowerment. S1 highlights how a low-code platform enables nontechnical employees to participate in development, lowering learning barriers, facilitating knowledge sharing, and supporting equal participation [32].

Frameworks in S7 and S9 address gender bias in career recommendations and recruitment [9, 24]. The career recommendation framework in S7 mitigates bias by neutralizing data representations and penalizing biased recommendations [9]. S9 examines a recruitment framework that excludes sensitive attributes such as gender and ethnicity, but finds that bias can persist when embedded in training data, emphasizing the need for advanced mitigation techniques [24].

Algorithms in S12 and S13 focus on reducing gender bias in language processing [3, 31]. The embedding algorithm in S12 removes bias by adjusting word associations while preserving meaningful relationships [31]. S13 refines this approach by identifying and neutralizing gendered subspaces, ensuring gender-neutral terms

are treated equally while maintaining distinctions for inherently gendered words [3].

These findings illustrate different approaches to mitigating bias in AI, demonstrating how tailored frameworks and algorithms contribute to more inclusive and equitable AI systems.

Table 6: Articles Focusing on Type of Artifact.

Study ID	Type of Artifact	Characteristics
S1	Low-Code Platform	Empowers non-technical users by reducing technical barriers, enabling diverse knowledge sharing, and fostering inclusive, equal opportunities for participation and contribution in development.
S7	Collaborative Filtering Framework	Corrects gender bias by adjusting initial data representations to be gender-neutral and by penalizing biased outcomes during the recommendation process, ensuring more balanced career suggestions across genders.
S9	Testbed Framework	Removes demographic bias in recruitment decisions by excluding sensitive information from decision-making.
S12	Embedding Algorithm	Reduces gender bias in word embeddings by removing gender-associated dimensions, creating more neutral semantic representations.
S13	Debiasing Algorithm	Utilizes hard debiasing techniques to reduce stereotypical gender associations in word embeddings.

5 Discussion

5.1 RQ1: How do recent studies address the empowerment of women in AI?

Recent years have shown a clear increase in publications on AI and empowerment, reflecting the rapid adoption of AI technologies [28]. This trend underlines the importance of ensuring that underrepresented groups are actively involved in shaping and applying these technologies.

While AI has been part of research and development for decades, the specific intersection between AI and empowerment has gained more attention in recent years. The launch of the UN's SDGs in 2015, particularly SDG 5 on gender equality, may have contributed to an increased research focus. However, societal movements like #MeToo, which gained global traction in 2017, did not immediately translate into a noticeable increase in academic publications, suggesting a potential lag between societal awareness and academic research output.

The geographical distribution of research is heavily skewed toward the Global North, with China being a notable exception from the Global South. This imbalance limits different perspectives, as AI applications and their impact on empowerment vary in economic, political, and technological contexts [6]. Studies from developed countries may not fully reflect the realities of underrepresented groups globally, reducing the generalizability of findings. The economic and infrastructural differences between regions highlight the need for more research that incorporates perspectives from diverse socio-political contexts.

Empowerment, although a central concept in many of the studies, is approached in notably different ways. For example, S2 views empowerment as building agency and resilience among women founders through interactive games, while S7 frames it as algorithmic fairness in career recommendation systems. S10 and S11 engage with empowerment through a feminist lens, urging for structural redesign in AI systems. These differing interpretations demonstrate a lack of consensus on what empowerment means in the context of AI, whether as a personal capacity, a technical intervention, or a systemic transformation.

Terms such as "bias," "gender", and "diversity" appear frequently, indicating an underlying motivation for empowerment. While many studies frame empowerment as a consequence of reducing bias, it should also be recognized as a proactive process that involves designing for inclusion, not just avoiding harm. AI's role extends beyond addressing existing inequalities to anticipating potential unintended consequences and shaping inclusive practices.

When comparing studies across the three thematic categories: AI and gender, engagement methods, and artifacts, a pattern emerges. Studies on AI and gender are often conceptual and advocate for inclusive design. Engagement-based studies show that well-designed learning environments or interventions can improve confidence and reshape perceptions. While artifact-centered studies mainly focus on bias reduction, but rarely engage with empowerment directly. This fragmentation in approach reflects the interdisciplinary nature of the field, but it also shows a lack of integration where technical, educational, and conceptual approaches often operate in parallel rather than in synergy.

As AI becomes increasingly integrated into daily life, the need to address gender equality within its design and application grows. Without careful consideration, AI risks reinforcing gender disparities. Encouraging greater participation of women in AI development is essential to ensuring that future AI systems prioritize equity and contribute to meaningful empowerment rather than perpetuating existing imbalances. The lack of diversity in development teams remains a persistent challenge, with studies emphasizing that homogeneous teams risk embedding existing biases into AI systems [4].

In summary, recent studies address women's empowerment in AI primarily through related themes such as inclusion and bias reduction, with empowerment often remaining an implicit or secondary focus.

5.2 RQ2: What are the characteristics of an AI artifact that facilitates the empowerment of women?

To better understand how AI can facilitate empowerment, this study examined key characteristics of AI artifacts, engagement methods, and their role in addressing gender disparities. However, a deeper analysis of the interactions between these categories reveals

opportunities for a more integrated approach to empowerment, one that bridges conceptual insight with practical interventions.

Studies addressing AI and gender emphasize the importance of transparent and inclusive designs [21, 22, 36]. These designs help identify and mitigate discrimination and support the goals of SDG 5 by detecting biases in data and decision-making. Additionally, diverse development teams are crucial for reducing bias, as a lack of diversity risks reinforcing existing inequalities [4, 36]. However, the studies rarely operationalize how such inclusive design practices can be translated into long-term empowerment, particularly in professional or educational settings.

AI systems often inherit biases embedded in training data, reinforcing gender inequalities [2]. To counter this, AI development must prioritize bias awareness and use representative datasets. Comprehensive and unbiased data collection helps prevent the reinforcement of traditional social roles, such as associating certain professions with specific genders. The inclusion of diverse development teams further enhances transparency and inclusivity [8, 32, 36, 38].

Methods to foster engagement with AI range from literacy courses [12] and summer camps [34] to robotic interactions [38] and interactive installations [8]. These approaches highlight the importance of educating individuals on AI's role in daily life and equipping them with practical tools for engagement [34, 38]. Many studies stress the need to make AI more accessible, reshaping perceptions and empowering underrepresented groups by providing a foundational skill set [14]. However, engagement methods often focus on introducing AI as a topic rather than on developing long-term strategies for sustained empowerment.

Findings on AI artifacts reinforce the importance of designing tools that actively reduce bias [9]. However, most studies focus on bias mitigation rather than directly fostering empowerment. This distinction is crucial, as bias reduction may be a prerequisite for empowerment, but it is not sufficient. The absence of artifacts specifically designed to empower underrepresented groups suggests a gap in research. Existing tools, such as low-code platforms, frameworks, and algorithms, aim to identify and address bias but lack explicit mechanisms for enhancing user empowerment. Future research should explore how AI can move beyond bias reduction to actively support empowerment by incorporating features that promote opportunities for marginalized groups.

In conclusion, empowering AI artifacts share features such as accessibility, bias mitigation, and inclusive design, though few explicitly aim to foster empowerment as a primary objective.

5.3 Limitations

This study has certain limitations that may affect the scope and generalizability of its findings.

One significant limitation is that only the Scopus database was used for the search of studies. While Scopus is a comprehensive source for academic research, incorporating additional databases such as the ACM Digital Library or IEEE Xplore could have enriched the scope of the study by identifying relevant studies that may not be indexed in Scopus. Expanding the search across multiple databases would have increased the likelihood of capturing a more diverse set of studies, particularly those focusing on AI applications in industry or specific regional contexts.

Another limitation lies in the search query used for this SLR. Only one search string was applied, which may have excluded studies that used alternate terminology, such as synonyms for "artifact" or "tool." Broadening the search query to include more aliases and alternate terms could have increased the inclusiveness of the dataset.

A third limitation is the reliance on a secondary source, where primary studies from one article [23] were used to identify studies from 2011 to 2021. Since this selection was based on the search conducted in that study, relevant articles may have been missed. While the search query aligns with this SLR, the lack of an independent search may have limited additional findings.

This SLR was conducted by a single author, which introduces potential bias in study selection and analysis. Although efforts were made to mitigate this through a double data search and periodic discussions with the second author, the absence of multiple reviewers in each phase of the SLR remains a limitation.

Finally, the geographic distribution of studies is skewed toward the Global North, with limited representation from the Global South. This underrepresentation restricts insights into regional and cultural differences in the intersection of AI and gender equality. Future research should expand geographic diversity to ensure a more comprehensive understanding.

6 Conclusion

This study investigated how AI research addresses the empowerment of women, through a systematic review of 14 peer-reviewed articles published between 2016 and 2024.

The findings show that while empowerment is frequently mentioned, few studies treat it as a central concept. Most research focuses on bias reduction, diversity, or inclusion, often indirectly contributing to empowerment through awareness-raising or participation in AI-related activities. Empowerment is interpreted in multiple ways, as increased confidence, access to tools, or ability to challenge bias, but these interpretations are rarely defined or evaluated systematically.

The analysis reveals that only a few AI artifacts are designed with empowerment in mind. Most tools focus on mitigating bias in data or decision-making processes, such as debiasing algorithms or fair recommendation systems. While some initiatives use interactive or educational technologies to engage underrepresented users, there is a lack of frameworks that explicitly link design features to empowerment outcomes.

The study also highlights a strong geographical imbalance, with most research coming from the Global North. This limits the applicability of findings across different cultural and economic contexts.

By identifying these gaps, this review contributes to a clearer understanding of how empowerment is currently addressed in AI research and practice. Future work will include an expanded search to incorporate additional studies. Building on previous research related to women in AI, the next step is to develop a research protocol, comprising both an interview guide and a questionnaire, to gain deeper insight into key factors and challenges. Empirical research is planned for this year to validate the findings of the systematic review and gather new input on barriers and success

factors for women in AI. A co-design process is also underway to develop an AI artifact that supports the empowerment of women in this field.

Future work will look into the relation between intersectionality and AI. As explained in section 2, gender is not the only diversity dimension. Future studies should look into the relation between AI and other dimensions of the intersectionality framework, such as ethnicity and disability.

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A Appendix A: Studies Included in the SLR

Study ID	Title
S1 [32]	How Low-Code Tools Contribute to Diversity, Equity, and Inclusion (DEI) in the Workplace: A Case Study of a Large Japanese Corporation
S2 [27]	Empowering Female Founders with AI and Play: Integration of a Large Language Model into a Serious Game with Player-Generated Content
S3 [8]	A Multimodal Installation Exploring Gender Bias in Artificial Intelligence
S4 [22]	Artificial Intelligence and Gender Equality: A Systematic Mapping Study
S5 [14]	Robots as My Future Colleagues: Changing Attitudes Toward Collaborative Robots by Means of Experience-Based Workshops
S6 [12]	Evaluation of an Artificial Intelligence Literacy Course for University Students with Diverse Study Backgrounds
S7 [9]	Debiasing Career Recommendations with Neural Fair Collaborative Filtering
S8 [38]	Boosting Robot Credibility and Challenging Gender Norms in Responding to Abusive Behaviour: A Case for Feminist Robots
S9 [24]	Bias in Multimodal AI: Testbed for Fair Automatic Recruitment
S10 [36]	Feminist AI: Can We Expect Our AI Systems to Become Feminist?
S11 [21]	A Gendered Perspective on Artificial Intelligence
S12 [31]	Biased Embeddings from Wild Data: Measuring, Understanding and Removing
S13 [3]	Man is to Computer Programmer as Woman is to Homemaker? Debiasing Word Embeddings
S14 [34]	Toward More Gender Diversity in CS through an Artificial Intelligence Summer Program for High School Girls